64. Which of the following are the public sector undertakings of the Government of India?

1. Balmer Lawrie and Company Ltd.

2. Dredging Corporation of India

3. Educational Consultants of India Ltd.

Select the correct answer using the code give below:

(a) 1 and 2 only (b) 2 and 3 only

(c) 1 and 3 only (d) 1, 2 and 3

Ans. (d) : Balmer Lawrie & Company Limited, Dredging Corporation of India and Educational Consultants of India Limited are Public Sector Undertakings of the Government of India.

PHYSICS

65. Consider the following statements:

1. The albedo of an object determines its visual brightness when viewed with reflected light.

2. The albedo of Mercury is much greater than the albedo of the Earth.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (a) : The albedo of an object will determine its visual brightness when viewed with reflected light. For example, the planets are viewed by reflected sunlight and their brightness depends upon the amount of light received from the sun and their albedo. The albedo of mercury is 0.12 while albedo of earth is 0.30. Therefore statement 2 is not correct.

66. How can the height of a person who is six feet tall be expressed (approximately) in nanometres?

(a) 183×10^6 nanometres

(b) 234×10^6 nanometres

(c) 183×10^7 nanometres

(d) 234×10^7 nanometres

Ans. (c) : 6 Feet=183×10^7 nanometres.

67. A person is sitting in a car which is at rest. The reaction from the road at each of the four wheels of the car is R. When the car runs on a straight level road, how will the reaction at either of the front wheels vary?

(a) It will be greater than R

(b) It will be less than R

(c) It will be equal to R

(d) It shall depend on the material of the road

Ans. (c) : Kinetic friction also known as dynamic or sliding friction occurs when a body is in relative motion with another with their surface in contact. It is the friction force that slows down and eventually stops a moving object. So, dynamic friction in any object is proportional to instant reaction and is less than static friction. When a car turns on a straight level road, the normal reaction either of the front wheels will be equal to R.

68. Consider the following statements:

1. A widely used musical scale called diatonics scale has seven frequencies.

2. The frequency of the note 'Sa' is 256 Hz and that of 'NI' is 512 Hz.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (a) : Tones with different frequencies are found in all types of popular music systems. Musical scale called diatonics scale has seven frequencies. Hence statement 1 is correct. The frequency of the note 'Sa' is 240 Hz and NI is 450 Hz. Therefore statement 2 is not correct.

69. Which one of the following laser types is used in a laser printer?

(a) Dye laser (b) Gas laser

(c) Semiconductor laser (d) Excimer laser

Ans. (c) : A semiconductor laser (LD) is a device that causes laser oscillation by flowing an electric current to semiconductor. It is used in Laser Printers.

70. Consider the following statements in respect of a jet engine and a rocket:

1. A Jet engine uses the surrounding air for its oxygen supply and so is unsuitable for motion in space.

2. A rocket carries its own supply of oxygen in the gas form and fuel.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (c) : A jet engine combines oxygen from the air with fuel at high temperature. There is usually a spark to ignite the fuel vapor, but once the engine begins turning, it will continue running until it runs out of fuel or air. So it is not suitable for motion in space. A rocket, in contrast, carries both fuel (which may be solid or liquid) and oxygen.

Chemistry

71. What is the pH level of blood of a normal person?

(a) 4.5-4.6 (b) 6.45-6.55

(c) 7.35-7.45 (d) 8.25-8.35

Ans. (c) : The normal pH value of blood is slightly basic. It ranges between 7.35 to. 7.45.

72. Which one of the following pairs of metals constitutes the lightest metal and the heaviest metal, respectively?

(a) Lithium and mercury

(b) Lithium and osmium

(c) Aluminium and osmium

(d) Aluminium and mercury

Ans. (b) : The lightest metal in the periodic table is lithium (Li) with atomic number 3 and density 0.53 kg/L. Lithium metal is extremely soft (and highly reactive) and so is unusable for many applications. Osmium is a hard metallic element which has the greatest density of all known elements. It is twice as heavy as lead, and has a specific gravity of 22.59.

73. Mixture of which one of the following pairs of gases is the cause of occurrence of most of the explosions in mines?

(a) Hydrogen and oxygen

(b) Oxygern and acetylene

(c) Methane and air

(d) Carbon dioxide and methane

Ans. (c) : Cause of blast in a mine in generally mixture of methane and air. CH_4 is the main gas exerted from a mine. When methane comes in contact with air, explosions take place.

74. Which one of the following is also called Stranger Gas?

(a) Argon (b) Neon

(c) Xenon (d) Nitrous oxide

Ans. (c) : Xenon is called the 'stranger' gas. The gas is very un-reactive and heavier than air, that is why named strange (in greek it means 'xenon').

75. What are Rubies and Sapphires chemically known as?

(a) Silicon dioxide

(b) Aluminum oxide

(c) Lead tetroxide

(d) Boron nitride

Ans. (b) : Rubies and Sapphires are scientifically the same stone, differing only in colour. Corundum, the predominating mineral of both, is composed of nearly pure alumina (Al2O3).The colouring substance which differentiates rubies and sapphires is believed to be chromium. Rubies and Sapphires are chemically known as aluminum oxide.

76. Who among the following discovered heavy water?

(a) Heinrich Hertz

(b) H. C. Urey

(c) G. Mendel

(d) Joseph Priestley

Ans. (b) : Heavy water (D_2O) was discovered by American scientist Harold Urey (H.C. Urey) in 1931. In all P.H.W.R. (Pressurized Heavy Water Reactor) D_2O (Heavy water) is used as a diluent or coolant.

BIOLOGY

77. Consider the following statements about probiotic food:

1. Probiotic food contains live bacteria which are considered beneficial to humans.

2. Probiotic food helps in maintaining gut flora.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (c) : Probiotics are the living microorganisms, healthy for host organism. Lactic acid bacteria, Bifido bacteria are common type of microbes used as probiotics. Certain yeast and Bacilli may also be useful. Probiotics are commonly consumed as a part of fermented food with specially added active live cultures such as yoghurt or as diatary supplements.

78. Among the following, which one lays eggs and does not produce young ones directly?

(a) Echidna (b) Kangaroo

(c) Porcupine (d) Whale

Ans. (a) : Echidna lays eggs, after which children develop from them. It is such a mammal that does not give babies directly.

79. The release of which one of the following into ponds and wells helps in controlling the mosquitoes?

(a) Crab (b) Dogfish

(c) Gambusia fish (d) Snail

Ans. (c) : Gambusia affinis is a larvivorous fish and is also known by the name mosquito fish. This fish is released into ponds and wells to control the mosquitoes by eating mosquito larvae, insects and crustaceans.

80. What is Bisphenol A (BPA)?

(a) A medical test for detecting cancer

(b) A test for testing the use of drugs to improve performance by athletes

(c) A chemical used for the development of food packaging materials

(d) A special type of alloy steel